

1. Query with learning outcomes of each study program.

```
(sp:StudyProgram)-[:HAS COURSE]->
```

(lo:LearningOutcome)

sp.programId AS ProgramId,

sp.title AS ProgramTitle,

collect(lo.text) AS LearningOutcomes

Result:

Business Informatics:

[illegible]

Digital Humanities:

[illegible]

2. Course found in multiple programs.

```
MATCH (p:StudyProgram)-[:HAS_COURSE]->(c:StudyCourse)
```

```
WITH c, count(p) AS programCount, collect(p.title) AS programs
```

```
WHERE programCount > 1
```

```
RETURN c.courseld AS courseld, c.title AS courseTitle, programCount, programs
```

Result:

1	"DE0751"	"e-Business Solutions"	["Business Informatics", "Digital Humanities"]
2	"DE0645"	"Portfolio Management Technologies"	["Business Informatics", "Digital Humanities"]
3	"DE0739"	"Knowledge Management Systems"	["Business Informatics", "Digital Humanities"]
4	"DE0729"	"Communication and Presentation Skills"	["Business Informatics", "Digital Humanities"]

3. Program with the highest number of courses.

```
MATCH (p:StudyProgram)-[:HAS_COURSE]->(c:StudyCourse)
```

```
RETURN p, count(c) AS courseCount
```

```
ORDER BY courseCount DESC
```

```
LIMIT 1
```

Result:

title
"Digital Humanities"

4. Course with the lowest number of outcomes.

```
MATCH (c:StudyCourse)
```

```
RETURN c, COUNT { (c)-[:HAS_OUTCOME]->() } AS outcomeCount
```

```
ORDER BY outcomeCount ASC
```

```
LIMIT 1
```

Result:

title

"Introduction to Engineering Sciences"
--

5. Topic repeating in multiple courses.

```
MATCH (c:StudyCourse)-[:HAS_TOPIC]->(t:CourseTopic)
```

```
WITH
```

```
t.title AS topicTitle,
```

```
collect(DISTINCT c.title) AS courses,
```

```
count(DISTINCT c) AS courseCount
```

```
WHERE courseCount > 1
```

```
RETURN topicTitle, courses, courseCount
```

```
ORDER BY courseCount DESC
```

Result:

```
"Assessment tests."
["Digital Language Learning Paradigm", "Digital Semantics and Pragmatics"]
```